Flow Matching and Diffusion Models

From transport PDEs to conditional velocity fields: a rigorous and unified course

Abstract

This course presents a self-contained, rigorously organized treatment of Flow Matching (FM) and its relationship to diffusion models. The logical chain is: (1) stochastic dynamics and the Fokker–Planck equation; (2) deterministic dynamics and the continuity equation with its mass conservation criterion; (3) the SDE/ODE bridge via the probability-flow ODE, reverse-time SDE, and marginal-preserving families; (4) the FM framework, which learns a velocity field whose ODE flow transports a source distribution to a target, using conditional paths and the marginalization trick for tractable training; (5) concrete constructions of probability paths and velocity fields, including the probability-path-first (Eulerian) and flow-first (Lagrangian) perspectives; (6) the full unification showing that Gaussian FM is equivalent to diffusion, that all four prediction parameterizations are interchangeable, and that all affine flows are equivalent up to time reparameterization.

Contents

I Transport PDEs and Dynamics **3**

1 Core objects and notation **3**

2 Stochastic dynamics and the Fokker–Planck equation **3**
2.1 Forward SDE . . . 3
2.2 The Fokker–Planck equation . . . 3
2.3 A fundamental score identity . . . 4

3 Deterministic dynamics and the continuity equation **4**
3.1 Flow maps and velocity fields . . . 4
3.2 The continuity equation . . . 5
3.3 Mass conservation criterion . . . 5

4 The SDE/ODE bridge **6**
4.1 Probability-flow ODE . . . 6
4.2 Reverse-time SDE . . . 7
4.3 A continuum of marginal-preserving SDEs . . . 7

II Flow Matching Framework **9**

5 From diffusion insights to the general framework **9**
5.1 Lesson from score-based diffusion . . . 9
5.2 The flow matching framework . . . 9
5.2.1 Step 1: Conditional path and its marginal densities . . . 10
5.2.2 Step 2: Velocity field . . . 10

5.2.3 Step 3: Learning via the conditional strategy . . . 10
5.3 The marginalization trick . . . 11
5.4 CFM and FM have the same gradient . . . 11

III Constructing Probability Paths and Velocity Fields **12**

6 Warm-up: Gaussian-to-Gaussian bridge **12**

7 Probability-path-first construction (Eulerian view) **12**
7.1 One-sided conditioning ($z = x_1 \sim p_{\text{tgt}}$) . . . 12
7.2 Two-sided conditioning ($z = (x_0, x_1)$) . . . 13
7.3 Gaussian FM = Diffusion . . . 13

8 Flow-first construction (Lagrangian view) **13**
8.1 Conditional affine flow . . . 14
8.2 One-sided ($z = x_1$) . . . 14
8.3 Two-sided ($z = (x_0, x_1)$) . . . 14
8.4 Recovery of the Gaussian bridge . . . 14

9 Path-first vs. flow-first: comparison **14**

IV Gaussian Paths, Scores, and Full Unification **15**

10 From forward SDEs to affine Gaussian schedules **15**
10.1 Linear SDE and its Gaussian conditional law . . . 15
10.2 Inverse identification: $(\alpha_t, \sigma_t) \to (f_t, g_t)$ . . . 16
10.3 VP and VE as special cases . . . 16

11 Score identities, four parameterizations, and the FM = Diffusion equivalence **16**
11.1 The affine Gaussian path . . . 16
11.2 Score marginalization for Gaussian mixtures . . . 17
11.3 The four prediction types and their algebraic equivalence . . . 17
11.4 The FM velocity is the PF-ODE drift: the complete equivalence . . . 18
11.5 PF-ODE in all four parameterizations . . . 19

12 All affine flows are equivalent **19**

V Synthesis **20**

13 The Fokker–Planck equation as the unifying principle **20**

A Derivation of weak forms from stochastic calculus **21**
A.1 ODE case . . . 21
A.2 SDE case . . . 21

Part I
Transport PDEs and Dynamics

1 Core objects and notation

Setting. We work in $\mathbb{R}^d$. All densities are with respect to the Lebesgue measure. We write p_t for the density of a random variable X_t at time t.

Definition 1.1 (Score function)**.** For a smooth, strictly positive density p, the *score* is

$$s(x) := \nabla_x \log p(x) = \frac{\nabla_x p(x)}{p(x)}.$$

For a time-indexed family $(p_t)_{t\in[0,T]}$, write $s(x,t) = \nabla_x \log p_t(x)$.

Definition 1.2 (Probability path)**.** A *probability path* is a family of densities $(p_t)_{t\in[0,1]}$ with prescribed boundary conditions, typically $p_0 = p_{\text{src}}$ (source) and $p_1 = p_{\text{tgt}}$ (target).

Central viewpoint

Both diffusion models and flow matching specify (explicitly or implicitly) a probability path $(p_t)_{t\in[0,1]}$ from a simple source p_0 to a data target p_1, and then construct dynamics — deterministic or stochastic — whose time marginals match this path. The entire FM–diffusion bridge rests on this principle.

2 Stochastic dynamics and the Fokker–Planck equation

We begin with the most general setting: stochastic differential equations and the PDE governing the evolution of their marginal densities.

2.1 Forward SDE

Definition 2.1 (Forward SDE)**.** Let $f : \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ be a drift and $g : [0,T] \to \mathbb{R}_{\geq 0}$ a scalar diffusion coefficient. A forward SDE is

$$\mathrm{d}X_t = f(X_t,t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \qquad X_0 \sim p_0, \tag{1}$$

where W_t is a standard d-dimensional Brownian motion. Under standard regularity (e.g., f Lipschitz in x, at most linear growth), (1) admits a unique strong solution with continuous sample paths.

2.2 The Fokker–Planck equation

The PDE governing the evolution of $p_t = \mathrm{Law}(X_t)$ is the Fokker–Planck equation. We state it first in weak form (the most natural formulation from the probabilistic viewpoint), then in strong form.

Definition 2.2 (Fokker–Planck equation — weak form)**.** Let $\varphi \in C_c^\infty(\mathbb{R}^d)$ be a test function. A family (p_t) satisfies the Fokker–Planck equation for the SDE (1) if, for all φ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} \varphi(x)\,p_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} \langle \nabla\varphi(x), f(x,t)\rangle\, p_t(x)\,\mathrm{d}x + \frac{g(t)^2}{2}\int_{\mathbb{R}^d} \Delta\varphi(x)\,p_t(x)\,\mathrm{d}x. \tag{2}$$

Derivation of the weak form. Apply Itô's formula to $\varphi(X_t)$ for the SDE (1):

$$\mathrm{d}\varphi(X_t) = \langle \nabla\varphi(X_t), f(X_t,t)\rangle\,\mathrm{d}t + g(t)\,\langle \nabla\varphi(X_t),\,\mathrm{d}W_t\rangle + \frac{g(t)^2}{2}\Delta\varphi(X_t)\,\mathrm{d}t.$$

Take expectations. The Itô integral $\int_0^t g(s)\,\langle \nabla\varphi(X_s),\,\mathrm{d}W_s\rangle$ is a martingale (under standard integrability), hence has zero expectation. Differentiating $\mathbb{E}[\varphi(X_t)] = \int \varphi(x)\,p_t(x)\,\mathrm{d}x$ yields (2). □

Definition 2.3 (Fokker–Planck equation — strong form)**.** If p_t is smooth and decays suitably at infinity, integration by parts in (2) yields the strong form:

$$\partial_t p_t = -\mathrm{div}(p_t f) + \frac{g(t)^2}{2}\,\Delta p_t. \tag{3}$$

2.3 A fundamental score identity

The following identity is used repeatedly throughout the course to trade Laplacians for score-weighted divergences.

Lemma 2.4 (Score–Laplacian identity)**.** *Let p be smooth and strictly positive, with score $s = \nabla \log p$. Then*

$$\mathrm{div}(p\,s) = \mathrm{div}(p\,\nabla\log p) = \mathrm{div}(\nabla p) = \Delta p. \tag{4}$$

Proof. $p\,\nabla\log p = p\cdot\frac{\nabla p}{p} = \nabla p$. Hence $\mathrm{div}(p\,s) = \mathrm{div}(\nabla p) = \Delta p$. □

Why scores enter all marginal equivalences

Identity (4) implies that the diffusion term $\frac{g^2}{2}\Delta p$ in Fokker–Planck can be rewritten as $\frac{g^2}{2}\mathrm{div}(p\,s)$. This is exactly what allows trading stochasticity (diffusion) for a deterministic correction involving the score, producing ODEs and families of SDEs with identical marginals.

3 Deterministic dynamics and the continuity equation

We now turn to the deterministic counterpart: ODE flows and the continuity equation.

3.1 Flow maps and velocity fields

Definition 3.1 (Flow map and pushforward)**.** A *flow map* is a family of measurable maps $\Psi_{s\to t}:\mathbb{R}^d\to\mathbb{R}^d$ indexed by $s,t\in[0,1]$, with $\Psi_{s\to s} = \mathrm{Id}$. Given $X_s\sim p_s$, define $X_t := \Psi_{s\to t}(X_s)$. The law of X_t is the *pushforward* $(\Psi_{s\to t})_{\#}p_s$, defined by

$$\int \varphi(x)\,(\Psi_{s\to t})_{\#}p_s(\,\mathrm{d}x) := \int \varphi(\Psi_{s\to t}(x_s))\,p_s(\,\mathrm{d}x_s), \quad \forall\,\varphi\in C_b(\mathbb{R}^d).$$

We write $p_t = (\Psi_{s\to t})_{\#}p_s$ when densities exist.

Definition 3.2 (Velocity field induced by a flow map)**.** Assume $\Psi_{0\to t}$ is differentiable in t and invertible for $t\in(0,1)$. The *velocity field* $v_t:\mathbb{R}^d\to\mathbb{R}^d$ is defined by

$$v_t(x) := \partial_t\Psi_{0\to t}\big(\Psi_{0\to t}^{-1}(x)\big). \tag{5}$$

Then trajectories satisfy the ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}X_t = v_t(X_t), \qquad X_0\sim p_0. \tag{6}$$

Flow map $\longleftrightarrow$ ODE: rigorous correspondence

If $\Psi_{0\to t}$ is invertible and differentiable, then v_t defined by (5) implies $X_t = \Psi_{0\to t}(X_0)$ solves $\dot{X}_t = v_t(X_t)$. Conversely, if v_t is (globally) Lipschitz in x (uniformly in t), then the ODE defines a unique flow map $\Psi_{0\to t}$ by $\Psi_{0\to t}(x_0) = X_t^{x_0}$.

3.2 The continuity equation

Definition 3.3 (Continuity equation — weak form)**.** A family of densities (p_t) satisfies the continuity equation with velocity field v_t if, for all $\varphi \in C_c^\infty(\mathbb{R}^d)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} \varphi(x)\, p_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} \langle \nabla\varphi(x), v_t(x)\rangle\; p_t(x)\,\mathrm{d}x. \tag{7}$$

Derivation of the weak form. If $\dot{X}_t = v_t(X_t)$ and $\varphi \in C_c^\infty$, then by the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi(X_t) = \langle \nabla\varphi(X_t), v_t(X_t)\rangle\,.$$

Taking expectations yields (7). □

Definition 3.4 (Continuity equation — strong form)**.** If p_t is smooth, integration by parts gives the strong form:

$$\partial_t p_t + \operatorname{div}(p_t\, v_t) = 0. \tag{8}$$

Remark 3.5 (Continuity = Fokker–Planck with $g \equiv 0$). Setting $g(t) \equiv 0$ in the Fokker–Planck equation (3) yields exactly the continuity equation (8) with $v_t = f(\cdot, t)$. The continuity equation is the deterministic analogue of Fokker–Planck.

3.3 Mass conservation criterion

Given a prescribed density path (p_t) and a velocity field (v_t), we want to know: *does the ODE flow induced by v_t produce exactly the densities p_t?*

Theorem 3.6 (Mass conservation criterion)**.** *Let $(p_t)_{t\in[0,1]}$ be a prescribed smooth density path and v_t a velocity field defining the ODE $\dot{X}_t = v_t(X_t)$ with flow map $\Psi_{0\to t}$. Let $p_t^{\mathrm{fwd}} := (\Psi_{0\to t})_\# p_0$ be the flow-induced density. Then, under standard regularity,*

$$p_t^{\mathrm{fwd}} = p_t \quad \textit{for all } t \in [0,1]$$

if and only if the pair (p_t, v_t) satisfies the continuity equation:

$$\partial_t p_t(x) + \nabla_x \cdot \big(p_t(x)\, v_t(x)\big) = 0, \qquad \forall\, t \in [0,1],\ x \in \mathbb{R}^d.$$

Proof sketch. ($\Rightarrow$): If $p_t^{\mathrm{fwd}} = p_t$, the weak continuity equation holds because, for any test function φ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(x)\, p_t(x)\,\mathrm{d}x = \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\varphi(X_t)] = \mathbb{E}[\langle \nabla\varphi(X_t), v_t(X_t)\rangle] = \int \langle \nabla\varphi(x), v_t(x)\rangle\; p_t(x)\,\mathrm{d}x.$$

($\Leftarrow$): If (p_t, v_t) satisfies the continuity equation, then p_t and p_t^{fwd} both solve the same linear PDE (the continuity equation) with the same initial condition p_0. Under uniqueness of weak solutions (guaranteed by Lipschitz regularity of v_t), they must coincide. □

Why the continuity equation is the underlying rule

The continuity equation is the single PDE criterion ensuring that a velocity field v_t generates the correct density path p_t. It is the *underlying rule* connecting:

- The probability-flow ODE in diffusion models (Section 4).
- The flow matching ODE (Section 5).
- The conditional-to-marginal construction (Proposition 5.1).

Remark 3.7 (Non-uniqueness of v_t for a given p_t). The continuity equation $\partial_t p_t + \mathrm{div}(p_t v_t) = 0$ is one scalar equation for d components of v_t: it is underdetermined. If v_t is a solution and $\tilde{v}_t$ satisfies $\mathrm{div}(p_t \tilde{v}_t) = 0$ (divergence-free with respect to p_t), then $v_t + \tilde{v}_t$ also solves the same equation. Flow matching resolves this non-uniqueness by *constructing* a particular v_t via conditional fields and regression, rather than solving the PDE directly.

4 The SDE/ODE bridge

We now establish the central PDE-level result: given a stochastic process with known marginals, one can construct deterministic and stochastic dynamics sharing the same marginals, using only the score function.

4.1 Probability-flow ODE

Theorem 4.1 (Probability-flow ODE)**.** *Let (p_t) be the marginal densities of the forward SDE* (1)*, assumed smooth and strictly positive for $t \in (0, T)$. Define the ODE*

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{X}_t = f(\tilde{X}_t, t) - \frac{g(t)^2}{2}\nabla_x \log p_t(\tilde{X}_t). \tag{9}$$

Then $\tilde{X}_t$ has the same time-marginal densities p_t as the SDE, under standard PDE uniqueness.

Proof. The SDE density satisfies Fokker–Planck:

$$\partial_t p_t = -\mathrm{div}(p_t f) + \frac{g(t)^2}{2}\Delta p_t.$$

The ODE (9) has drift $\tilde{f} = f - \frac{g^2}{2}s$ (where $s = \nabla \log p_t$), so its density satisfies the continuity equation:

$$\partial_t \tilde{p}_t = -\mathrm{div}\Big(\tilde{p}_t\big(f - \tfrac{g^2}{2}s\big)\Big) = -\mathrm{div}(\tilde{p}_t f) + \frac{g^2}{2}\mathrm{div}(\tilde{p}_t s).$$

By the score–Laplacian identity (4), $\mathrm{div}(p_t s) = \Delta p_t$. Provided $\tilde{p}_t = p_t$, the right-hand side becomes $-\mathrm{div}(p_t f) + \frac{g^2}{2}\Delta p_t$, which equals $\partial_t p_t$ by Fokker–Planck. Hence p_t satisfies the continuity equation for $\tilde{f}$, and by uniqueness, $\tilde{p}_t = p_t$. □

The probability-flow ODE in words

The PF-ODE removes all stochasticity from the forward SDE while preserving marginal distributions. The "missing" diffusion is exactly compensated by a deterministic score correction $-\frac{g^2}{2}\nabla \log p_t$. This is possible because $\mathrm{div}(p\,\nabla \log p) = \Delta p$: the score encodes precisely the density-level effect of diffusion.

4.2 Reverse-time SDE

Theorem 4.2 (Reverse-time SDE)**.** *Let (p_t) be the marginal densities of the forward SDE* (1) *with $X_0 \sim p_{\text{data}}$. Define the reversed-time density $\tilde{p}_\tau(x) := p_{T-\tau}(x)$ for $\tau \in [0, T]$. The reverse-time SDE*

$$\mathrm{d}\bar{X}_\tau = \Big[-f(\bar{X}_\tau, T-\tau) + g(T-\tau)^2 \nabla \log p_{T-\tau}(\bar{X}_\tau) \Big] \mathrm{d}\tau + g(T-\tau)\, \mathrm{d}\bar{W}_\tau, \tag{10}$$

with $\bar{X}_0 \sim p_T$, has marginal densities $\tilde{p}_\tau = p_{T-\tau}$.

Proof. Since $\tilde{p}_\tau = p_{T-\tau}$, we have $\partial_\tau \tilde{p}_\tau = -\partial_t p_t\big|_{t=T-\tau}$. Using the forward Fokker–Planck:

$$\partial_\tau \tilde{p}_\tau = \operatorname{div}(\tilde{p}_\tau\, f(\cdot, T-\tau)) - \frac{g(T-\tau)^2}{2} \Delta \tilde{p}_\tau.$$

On the other hand, if (10) holds with drift $\tilde{h}$, then $\tilde{p}_\tau$ must satisfy

$$\partial_\tau \tilde{p}_\tau = -\operatorname{div}(\tilde{p}_\tau\, \tilde{h}) + \frac{g(T-\tau)^2}{2} \Delta \tilde{p}_\tau.$$

Equating and using $\Delta \tilde{p}_\tau = \operatorname{div}(\tilde{p}_\tau \nabla \log \tilde{p}_\tau)$ yields

$$\tilde{h}(x, \tau) = -f(x, T-\tau) + g(T-\tau)^2 \nabla \log p_{T-\tau}(x),$$

which is the drift in (10). □

Remark 4.3. In the original time variable t (decreasing from T to 0), the reverse SDE reads:

$$\mathrm{d}X_t = \big[f(X_t, t) - g(t)^2 \nabla \log p_t(X_t) \big] \mathrm{d}t + g(t)\, \mathrm{d}\bar{W}_t,$$

where the correction $-g^2 s$ appears with coefficient 1 (not $\frac{1}{2}$ as in the PF-ODE) because the diffusion term contributes an additional $+\frac{g^2}{2} s$ to the Fokker–Planck balance.

4.3 A continuum of marginal-preserving SDEs

Theorem 4.4 (Marginal-preserving SDE family)**.** *Assume (p_t) is smooth and strictly positive and satisfies the continuity equation with velocity field v_t:*

$$\partial_t p_t + \operatorname{div}(p_t\, v_t) = 0.$$

Let $\gamma(t) \geq 0$ be any scalar schedule. Define the SDE

$$\mathrm{d}X_t = b(X_t, t)\, \mathrm{d}t + \gamma(t)\, \mathrm{d}B_t, \qquad X_0 \sim p_0, \tag{11}$$

with drift

$$b(x, t) := v_t(x) + \frac{\gamma(t)^2}{2} \nabla \log p_t(x). \tag{12}$$

Then (p_t) also satisfies the Fokker–Planck equation for (11)*. In particular,* $\operatorname{Law}(X_t) = p_t$ *for all t.*

Proof. The Fokker–Planck right-hand side for (11) is:

$$-\operatorname{div}(p_t b) + \frac{\gamma^2}{2} \Delta p_t = -\operatorname{div}(p_t v_t) - \frac{\gamma^2}{2} \operatorname{div}(p_t\, s) + \frac{\gamma^2}{2} \Delta p_t.$$

By $\operatorname{div}(p_t s) = \Delta p_t$, the last two terms cancel, leaving $-\operatorname{div}(p_t v_t) = \partial_t p_t$. □

Three key dynamics from one density path

Given a density path (p_t) with score $s(\cdot, t) = \nabla \log p_t$:

1. **ODE** ($\gamma \equiv 0$): $\mathrm{d}X_t = v_t(X_t)\,\mathrm{d}t$ (deterministic, PF-ODE)
2. **Reverse SDE** ($\gamma = g$): the canonical Anderson reversal.
3. **General SDE** (arbitrary γ): interpolates between deterministic and stochastic.

All three share the same marginal densities p_t, by Theorem 4.4.

Part II
Flow Matching Framework

5 From diffusion insights to the general framework

5.1 Lesson from score-based diffusion

Before defining flow matching in full generality, we extract the key structural insights from diffusion models (Chapter 4). The construction of a diffusion-based generative model follows three steps:

Step 1: A conditional path and its marginal densities. A diffusion model defines a family of densities $(p_t)_{t\in[0,1]}$ via a forward conditional distribution

$$p_t(x_t \mid x_0) = \mathcal{N}(x_t;\, \alpha_t x_0,\, \sigma_t^2 \mathrm{Id}), \qquad x_0 \sim p_{\mathrm{data}}, \tag{13}$$

which induces the marginal density $p_t(x_t) = \int p_t(x_t \mid x_0)\, p_{\mathrm{data}}(x_0)\, \mathrm{d}x_0$. The increasing variance σ_t^2 drives p_t toward a Gaussian prior.

Step 2: A velocity field from the Fokker–Planck equation. The marginal p_t evolves according to a velocity field $v_t(x) = f(t)x - \frac{1}{2}g^2(t)\nabla \log p_t(x)$, derived from the probability-flow ODE (Theorem 4.1). The ODE $\dot{x}(t) = v_t(x(t))$ transports samples along (p_t). The underlying PDE ensuring this alignment is the continuity equation (Theorem 3.6).

Step 3: Learning via the conditional strategy. Since $\nabla \log p_t(x_t)$ is intractable, we exploit the tractable conditional score $\nabla \log p_t(x_t \mid x_0)$. The key identity

$$\nabla \log p_t(x_t) = \mathbb{E}_{x_0 \sim p(\cdot \mid x_t)}\big[\nabla \log p_t(x_t \mid x_0)\big]$$

shows that marginal scores are conditional expectations of conditional scores. Training with conditional targets (denoising score matching) differs from the marginal objective only by a θ-independent constant.

The three-step pattern

1. **Define** a conditional path $p_t(\cdot \mid z)$ and its marginal $p_t = \int p_t(\cdot \mid z)\pi(z)\, \mathrm{d}z$.
2. **Identify** a velocity field v_t such that (p_t, v_t) satisfies the continuity equation.
3. **Learn** v_t by regressing against tractable conditional targets $v_t(\cdot \mid z)$.

Flow matching generalizes this pattern to *arbitrary* source and target distributions.

5.2 The flow matching framework

We now present the FM framework in its full generality, following the same three-step structure.

Setup. Let p_{src} and p_{tgt} be two distributions on $\mathbb{R}^d$ (source and target), both easy to sample from. We set $p_0 = p_{\mathrm{src}}$ and $p_1 = p_{\mathrm{tgt}}$. The generation task (source = Gaussian prior, target = data) is a special case.

5.2.1 Step 1: Conditional path and its marginal densities

FM implicitly defines a continuous family $(p_t)_{t\in[0,1]}$ interpolating between the endpoints. Each marginal p_t is expressed via a latent variable $z \sim \pi(z)$ and a conditional distribution $p_t(\cdot \mid z)$:

$$p_t(x_t) = \int p_t(x_t \mid z)\, \pi(z)\, \mathrm{d}z, \tag{14}$$

with $(\pi, \{p_t(\cdot \mid z)\})$ chosen to satisfy the boundary conditions $p_0 = p_{\mathrm{src}}$, $p_1 = p_{\mathrm{tgt}}$.

Common choices for the latent variable z:

- **Two-sided conditioning**: $z = (x_0, x_1)$ with $\pi = p_{\mathrm{src}}(x_0)\, p_{\mathrm{tgt}}(x_1)$.
- **One-sided conditioning**: $z = x_1 \sim p_{\mathrm{tgt}}$ (or $z = x_0 \sim p_{\mathrm{src}}$), recovering diffusion-like setups when p_{src} is Gaussian.

5.2.2 Step 2: Velocity field

The goal is to find a velocity field $v_t(x)$ such that the ODE

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} = v_t(x(t)), \qquad x(0) \sim p_{\mathrm{src}}, \tag{15}$$

produces marginal distributions matching p_t at each time t. By Theorem 3.6, this requires (p_t, v_t) to satisfy the continuity equation:

$$\partial_t p_t(x) + \nabla_x \cdot \big(p_t(x)\, v_t(x)\big) = 0. \tag{16}$$

As noted in Section 3.3, v_t is not unique for a given p_t. FM seeks one particular solution by marginalization of conditional velocity fields.

5.2.3 Step 3: Learning via the conditional strategy

Conditional velocity field. For each z, we define (or derive) a conditional velocity field $v_t(\cdot \mid z)$ such that the conditional ODE

$$\dot{X}_t = v_t(X_t \mid z), \qquad X_0 \sim p_0(\cdot \mid z),$$

generates $p_t(\cdot \mid z)$.

Marginal velocity by marginalization. The marginal velocity field is defined as

$$v_t(x) \coloneqq \int v_t(x \mid z)\, \frac{p_t(x \mid z)\, \pi(z)}{p_t(x)}\, \mathrm{d}z = \mathbb{E}_{z\sim p(\cdot \mid x_t = x)}\big[v_t(x \mid z)\big]. \tag{17}$$

FM and CFM losses. The FM loss is

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t,\, x_t \sim p_t}\big[\, \|v_\theta(x_t, t) - v_t(x_t)\|^2 \,\big],$$

which is intractable because v_t involves an expectation over the posterior $p(z \mid x_t)$. The *conditional* FM loss is

$$\mathcal{L}_{\mathrm{CFM}}(\theta) = \mathbb{E}_{t,\, z\sim\pi,\, x_t \sim p_t(\cdot \mid z)}\big[\, \|v_\theta(x_t, t) - v_t(x_t \mid z)\|^2 \,\big]. \tag{18}$$

We prove below that these two losses have the same gradient and the same minimizer.

5.3 The marginalization trick

Proposition 5.1 (Marginal velocity field generates the marginal path)**.** *If each conditional velocity field $v_t(\cdot \mid z)$ generates $p_t(\cdot \mid z)$ (i.e., satisfies $\partial_t p_t(\cdot \mid z) + \operatorname{div}(p_t(\cdot \mid z)\, v_t(\cdot \mid z)) = 0$), then the marginal velocity field v_t defined by* (17) *generates the marginal path p_t.*

Proof. Fix $\varphi \in C_c^\infty(\mathbb{R}^d)$. For each z, the conditional continuity equation in weak form gives:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int \varphi(x)\, p_t(x \mid z)\, \mathrm{d}x = \int \langle \nabla\varphi(x), v_t(x \mid z) \rangle\; p_t(x \mid z)\, \mathrm{d}x.$$

Multiply by $\pi(z)$ and integrate over z:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int \varphi(x) \underbrace{\Big(\int p_t(x \mid z) \pi(z)\, \mathrm{d}z \Big)}_{p_t(x)} \mathrm{d}x = \int \Bigg\langle \nabla\varphi(x), \underbrace{\int v_t(x \mid z)\, p_t(x \mid z)\, \pi(z)\, \mathrm{d}z}_{v_t(x)\, p_t(x)} \Bigg\rangle \mathrm{d}x.$$

This is exactly the weak continuity equation for (p_t, v_t). By Theorem 3.6, the ODE flow of v_t generates p_t. □

The marginalization trick in one line

$$v_t(x) = \mathbb{E}\big[v_t(x \mid Z) \,\big|\, X_t = x \big].$$

This identity converts a family of tractable conditional velocity fields into a single marginal velocity field that generates the correct marginal density path.

5.4 CFM and FM have the same gradient

Lemma 5.2 (Variance decomposition for regression)**.** *Let A be square-integrable and X any random variable. For any measurable predictor $f(X)$,*

$$\mathbb{E}\big[\, \|f(X) - A\|^2 \,\big] = \mathbb{E}\big[\, \|f(X) - \mathbb{E}[A \mid X]\|^2 \,\big] + \mathbb{E}\big[\, \|A - \mathbb{E}[A \mid X]\|^2 \,\big].$$

Proof. Expand $\|f - A\|^2 = \|(f - \mathbb{E}[A \mid X]) + (\mathbb{E}[A \mid X] - A)\|^2$ and note that the cross term has zero expectation because $\mathbb{E}[A - \mathbb{E}[A \mid X] \mid X] = 0$. □

Theorem 5.3 (Equivalence of CFM and FM)**.** *Let $v_t(x) = \mathbb{E}[v_t(x \mid Z) \mid X_t = x]$ be the marginal velocity. Then*

$$\mathcal{L}_{\mathrm{CFM}}(\theta) = \mathcal{L}_{\mathrm{FM}}(\theta) + C,$$

where C does not depend on θ. In particular, $\nabla_\theta \mathcal{L}_{\mathrm{CFM}} = \nabla_\theta \mathcal{L}_{\mathrm{FM}}$, and the unique minimizer satisfies

$$v_\theta^*(x_t, t) = v_t(x_t) = \mathbb{E}\big[v_t(x_t \mid Z) \,\big|\, X_t = x_t \big].$$

Proof. Apply Lemma 5.2 with $A = v_t(X_t \mid Z)$, $X = X_t$, and $f = v_\theta(\cdot, t)$. Then $\mathbb{E}[A \mid X_t] = v_t(X_t)$ by definition (17), and the residual term $\mathbb{E}[\|A - \mathbb{E}[A \mid X_t]\|^2]$ is θ-independent. □

Practical meaning

Training with tractable conditional targets $v_t(\cdot \mid z)$ (CFM) is not an approximation. It is an exactly justified surrogate: same gradients, same minimizer as training with the intractable marginal target v_t. This is the same principle as denoising score matching in diffusion models.

Part III
Constructing Probability Paths and Velocity Fields

6 Warm-up: Gaussian-to-Gaussian bridge

When both p_{src} and p_{tgt} are Gaussian, the marginal density and velocity can be computed analytically. This serves as a template for the general conditional construction.

Consider the interpolated Gaussian path:

$$p_t(x) = \mathcal{N}\big(x;\, \mu(t),\, \sigma(t)^2\mathrm{Id}\big), \tag{19}$$

with endpoints $p_0 = \mathcal{N}(\mu(0), \sigma(0)^2\mathrm{Id})$ and $p_1 = \mathcal{N}(\mu(1), \sigma(1)^2\mathrm{Id})$.

A natural affine flow map realizing this path is:

$$\Psi_{0\to t}(x) = \mu(t) + \frac{\sigma(t)}{\sigma(0)}\big(x - \mu(0)\big). \tag{20}$$

If $x \sim p_0$, then $\Psi_{0\to t}(x) \sim \mathcal{N}(\mu(t), \sigma(t)^2\mathrm{Id}) = p_t$.

Proposition 6.1 (Velocity field for a Gaussian density path)**.** *The velocity field generating the flow* (20) *is:*

$$v_t(x) = \frac{\sigma'(t)}{\sigma(t)}\big(x - \mu(t)\big) + \mu'(t). \tag{21}$$

Proof. Differentiate $\Psi_{0\to t}(y)$ in t: $\Psi'_{0\to t}(y) = \mu'(t) + \frac{\sigma'(t)}{\sigma(0)}(y - \mu(0))$. The inverse is $y = \Psi_{t\to 0}(x) = \mu(0) + \frac{\sigma(0)}{\sigma(t)}(x - \mu(t))$. Substituting: $v_t(x) = \Psi'_{0\to t}(\Psi_{t\to 0}(x)) = \mu'(t) + \frac{\sigma'(t)}{\sigma(t)}(x - \mu(t))$. □

Remark 6.2 (Two perspectives on uniqueness). For a *fixed flow map* $\Psi_{0\to t}$, the velocity is uniquely determined by $v_t = \partial_t \Psi_{0\to t} \circ \Psi^{-1}_{0\to t}$. For a *fixed density path* p_t (without fixing $\Psi_{0\to t}$), the velocity is not unique: the continuity equation admits infinitely many solutions. This distinction precisely characterizes the difference between the flow-first and probability-path-first perspectives below.

7 Probability-path-first construction (Eulerian view)

Principle: Choose a conditional probability path $p_t(\cdot \mid z)$ first, then derive the conditional velocity $v_t(\cdot \mid z)$.

7.1 One-sided conditioning ($z = x_1 \sim p_{\text{tgt}}$)

Fix $x_1 \sim p_{\text{tgt}}$. Define the conditional Gaussian path:

$$p_t(x_t \mid x_1) = \mathcal{N}\big(x_t;\, b_t x_1,\, a_t^2\mathrm{Id}\big), \tag{22}$$

with boundary conditions $a_0 = 1$, $b_0 = 0$ and $a_1 = 0$, $b_1 = 1$ (the limit $a_1 \to 0$ concentrates $p_1(\cdot \mid x_1)$ at δ_{x_1}).

At the boundaries: $p_0(\cdot \mid x_1) = \mathcal{N}(0, \mathrm{Id})$ (independent of x_1) and $p_1(\cdot \mid x_1) = \delta(\cdot - x_1)$. Marginalizing: $p_0 = \mathcal{N}(0, \mathrm{Id}) = p_{\text{src}}$ and $p_1 = p_{\text{tgt}}$.

Applying Proposition 6.1 to the conditional Gaussian path (22) (with $\mu(t) = b_t x_1$ and $\sigma(t) = a_t$):

Proposition 7.1 (Conditional velocity for one-sided path-first construction)**.** *For $t \in (0, 1)$ with $a_t > 0$:*

$$v_t(x \mid x_1) = b'_t x_1 + \frac{a'_t}{a_t}(x - b_t x_1). \tag{23}$$

CFM loss. Sampling $x_t = a_t\varepsilon + b_t x_1$ with $\varepsilon \sim \mathcal{N}(0, \mathrm{Id})$, the CFM loss becomes:

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t,\, x_1 \sim p_{\mathrm{tgt}},\, x_t \sim p_t(\cdot|x_1)}\left[\left\| v_\theta(x_t, t) - \left[b_t' x_1 + \frac{a_t'}{a_t}(x_t - b_t x_1)\right]\right\|^2\right]. \tag{24}$$

On paired samples. Writing $x_0 = \varepsilon$ and $x_t = a_t x_0 + b_t x_1$, the conditional velocity simplifies to $v_t(x_t \mid x_1) = a_t' x_0 + b_t' x_1$, which is the time derivative of the interpolation x_t.

7.2 Two-sided conditioning ($z = (x_0, x_1)$)

Let $z = (x_0, x_1)$ with $x_0 \sim p_{\mathrm{src}}$, $x_1 \sim p_{\mathrm{tgt}}$ independently. Define:

$$p_t(x_t \mid x_0, x_1) = \mathcal{N}\big(x_t;\, a_t x_0 + b_t x_1,\, \sigma^2 \mathrm{Id}\big), \tag{25}$$

with $a_0 = 1$, $b_0 = 0$, $a_1 = 0$, $b_1 = 1$. In the deterministic limit $\sigma = 0$: $p_t(\cdot \mid x_0, x_1) = \delta(\cdot - (a_t x_0 + b_t x_1))$.

Proposition 7.2 (Conditional velocity for two-sided construction)**.**

$$v_t(x \mid x_0, x_1) = a_t' x_0 + b_t' x_1. \tag{26}$$

The CFM loss (with $\sigma = 0$, so $x_t = a_t x_0 + b_t x_1$):

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t,\, x_0 \sim p_{\mathrm{src}},\, x_1 \sim p_{\mathrm{tgt}}}\big[\, \big\| v_\theta(x_t, t) - (a_t' x_0 + b_t' x_1)\big\|^2 \,\big].$$

Remark 7.3 (Same minimizer for one-sided and two-sided). On paired samples (x_0, x_1) with $x_t = a_t x_0 + b_t x_1$, both the one-sided target $v_t(x_t \mid x_1) = a_t' x_0 + b_t' x_1$ and the two-sided target coincide. The optimal velocity field is:

$$v^*(x, t) = \mathbb{E}\big[a_t' X_0 + b_t' X_1 \,\big|\, X_t = x\big].$$

7.3 Gaussian FM = Diffusion

When $p_{\mathrm{src}} = \mathcal{N}(0, \mathrm{Id})$, the one-sided conditioning on $z = x_1 \sim p_{\mathrm{data}}$ with Gaussian conditional paths is called *Gaussian FM*. Under the relabeling $a_t \mapsto \sigma_t$, $b_t \mapsto \alpha_t$ (matching diffusion notation), this is exactly the diffusion model trained to predict the velocity.

The canonical choice $a_t = 1 - t$, $b_t = t$ (equivalently $\sigma_t = 1 - t$, $\alpha_t = t$) gives the FM/Rectified Flow schedule with CFM target $v_t(x_t \mid x_1) = x_1 - x_0$.

	VE	**VP**	**FM/RF**	**Trigonometric**
a_t (source coeff.)	a_t	$\sqrt{1 - b_t^2}$	$1 - t$	$\cos(\frac{\pi}{2}t)$
b_t (target coeff.)	1	b_t	t	$\sin(\frac{\pi}{2}t)$
a_0, b_0	$\cdot$, 1	$\cdot$, 1	1, 0	1, 0
a_1, b_1	a_1, 1	0, $\cdot$	0, 1	0, 1
p_{src}	$\mathcal{N}(0, a_1^2 \mathrm{Id})$	$\mathcal{N}(0, \mathrm{Id})$	$\mathcal{N}(0, \mathrm{Id})$	$\mathcal{N}(0, \mathrm{Id})$

Table 1: Common interpolant schedules in the FM convention $x_t = a_t x_0 + b_t x_1$, where $x_0 \sim p_{\mathrm{src}}$ and $x_1 \sim p_{\mathrm{tgt}}$.

8 Flow-first construction (Lagrangian view)

Principle: Specify a conditional flow map $\Psi_{0\to t}(\cdot \mid z)$ directly, then derive the conditional velocity by time-differentiation along trajectories.

8.1 Conditional affine flow

Fix $z \sim \pi$ and define the time-varying conditional affine flow:

$$\Psi_{0\to t}(x_0 \mid z) \coloneqq \mu_t(z) + A_t(z)\, x_0, \tag{27}$$

where $\mu_t(z) \in \mathbb{R}^d$ and $A_t(z) \in \mathbb{R}^{d\times d}$ is invertible for $t \in (0,1)$. Boundary: $A_0(z) = \mathrm{Id}$, $\mu_0(z) = 0$ (recovers p_{src} at $t = 0$).

Induced conditional path. $p_t(\cdot \mid z) = (\Psi_{0\to t}(\cdot \mid z))_{\#} p_{\mathrm{src}}$. Sampling: draw $x_0 \sim p_{\mathrm{src}}$, set $x_t = \mu_t(z) + A_t(z)\, x_0$.

Derived conditional velocity. The conditional velocity is obtained by differentiating the flow map along its trajectory:

$$v_t(x \mid z) = \frac{\mathrm{d}}{\mathrm{d}t}\Psi_{0\to t}\big(\Psi_{t\to 0}(x \mid z) \mid z\big). \tag{28}$$

Since $x = \mu_t(z) + A_t(z)\, x_0$, we have $x_0 = A_t(z)^{-1}(x - \mu_t(z))$, giving:

$$v_t(x \mid z) = \mu_t'(z) + A_t'(z)\, A_t(z)^{-1}\big(x - \mu_t(z)\big). \tag{29}$$

8.2 One-sided ($z = x_1$)

Set $\mu_t(z) = b_t x_1$ and $A_t(z) = a_t \mathrm{Id}$ with $a_0 = 1$, $b_0 = 0$, $a_1 = 0$, $b_1 = 1$:

$$x_t = a_t x_0 + b_t x_1, \qquad v_t(x \mid x_1) = b_t' x_1 + \frac{a_t'}{a_t}(x - b_t x_1).$$

On paired samples: $v_t(x_t \mid x_1) = a_t' x_0 + b_t' x_1$, recovering the same CFM target as the path-first approach.

8.3 Two-sided ($z = (x_0, x_1)$)

With $\mu_t(x_0, x_1) = b_t x_1$ and $A_t = a_t \mathrm{Id}$:

$$x_t = a_t x_0 + b_t x_1, \qquad v_t(x_t \mid x_0, x_1) = a_t' x_0 + b_t' x_1.$$

8.4 Recovery of the Gaussian bridge

If μ_t is independent of z (denoted $\mu(t)$) and $A_t = \frac{\sigma(t)}{\sigma(0)}\mathrm{Id}$, then

$$\Psi_{0\to t}(x_0) = \mu(t) + \frac{\sigma(t)}{\sigma(0)}(x_0 - \mu(0) + \mu(0)) = \mu(t) + \frac{\sigma(t)}{\sigma(0)}(x_0 - \mu(0)),$$

and $v_t(x) = \mu'(t) + \frac{\sigma'(t)}{\sigma(t)}(x - \mu(t))$, recovering the Gaussian bridge velocity of Proposition 6.1.

9 Path-first vs. flow-first: comparison

Takeaway

Path-first is natural when conditional paths admit closed-form velocities (e.g., Gaussians). Flow-first is natural when you have strong structural priors on trajectories (e.g., affine maps, geodesics on manifolds). Both yield equivalent transport under regularity and lead to the same CFM training objective.

Axis	Path-First (Eulerian)	Flow-First (Lagrangian)
Given	Conditional density path $p_t(\cdot \mid z)$	Conditional flow map $\Psi_{0\to t}(\cdot \mid z)$
Derive	Velocity $v_t(\cdot \mid z)$ by solving the continuity equation for $p_t(\cdot \mid z)$	Velocity $v_t(x \mid z) = \partial_t \Psi_{0\to t}(\Psi_{t\to 0}(x \mid z) \mid z)$
Uniqueness of v_t	Not unique for a given $p_t(\cdot \mid z)$ (one scalar PDE, d unknowns)	Unique given $\Psi_{0\to t}(\cdot \mid z)$
Realizability	Must verify the constructed v_t satisfies the continuity equation	Holds by construction: $p_t(\cdot \mid z) = (\Psi_{0\to t})_{\#} p_0(\cdot \mid z)$
Closed form	Convenient when $p_t(\cdot \mid z)$ is Gaussian / exponential family	Convenient when $\Psi_{0\to t}$ has structure (affine, geodesic)
Preferred use	Diffusion-style constructions; analytic conditional Gaussians	Structural priors on trajectories; boundary control; manifolds

Table 2: Comparison of the two complementary perspectives for constructing conditional paths and velocities.

Part IV
Gaussian Paths, Scores, and Full Unification

In Parts II and III we developed the FM framework in full generality: conditional paths, conditional velocity fields, the marginalization trick, and CFM training. We now specialize to *affine Gaussian* paths, the setting where FM and diffusion models coincide. The goal of this part is threefold:

1. Show how forward diffusion SDEs produce the affine Gaussian conditional structure.
2. Derive, in a single pass, all score-based identities and the algebraic equivalences among the four prediction types (ε, x, score, v).
3. State the FM = Diffusion equivalence as a clean theorem.

10 From forward SDEs to affine Gaussian schedules

This section answers a natural question: *where do the schedules* (α_t, σ_t) *come from?* The answer: they are the mean and noise coefficients of the Gaussian conditional law of a linear forward SDE.

10.1 Linear SDE and its Gaussian conditional law

Consider the linear SDE:

$$\mathrm{d}Z_t = f_t Z_t \,\mathrm{d}t + g_t \,\mathrm{d}B_t, \qquad Z_0 \sim p_{\text{data}}, \qquad t \in [0, T], \tag{30}$$

with scalar $f_t \in \mathbb{R}$ and $g_t \geq 0$.

Definition 10.1 (Mean coefficient and noise scale)**.**

$$\alpha_t \coloneqq \exp\Big(\int_0^t f_s \,\mathrm{d}s\Big), \qquad \sigma_t^2 \coloneqq \alpha_t^2 \int_0^t g_s^2 \,\alpha_s^{-2} \,\mathrm{d}s. \tag{31}$$

Lemma 10.2 (Gaussian conditional law)**.** *The unique strong solution of* (30) *is* $Z_t = \alpha_t Z_0 + \alpha_t \int_0^t \alpha_s^{-1} g_s \,\mathrm{d}B_s$*. Conditionally on* Z_0*:*

$$Z_t \mid Z_0 \overset{d}{=} \alpha_t Z_0 + \sigma_t \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathrm{Id}). \tag{32}$$

Moreover, σ_t^2 satisfies $\dot{\sigma}_t^2 = 2f_t\sigma_t^2 + g_t^2$, $\sigma_0^2 = 0$.

Proof. Apply Itô to $\alpha_t^{-1}Z_t$ using $\dot{\alpha}_t = f_t\alpha_t$: $\mathrm{d}(\alpha_t^{-1}Z_t) = \alpha_t^{-1}g_t\,\mathrm{d}B_t$. Integrate and use Itô isometry to compute the variance. □

The Gaussian conditional structure

Every linear SDE produces Gaussian conditionals $Z_t \mid Z_0 \sim \mathcal{N}(\alpha_t Z_0,\, \sigma_t^2\mathrm{Id})$. This is the same affine Gaussian structure that FM uses for conditional paths (Section 7). The forward diffusion SDE is thus one *origin story* for the schedules (α_t, σ_t); FM can also choose them freely.

10.2 Inverse identification: $(\alpha_t, \sigma_t) \to (f_t, g_t)$

Proposition 10.3. *If $\alpha_t > 0$ and $\sigma_t > 0$ are differentiable:*

$$f_t = \frac{\dot{\alpha}_t}{\alpha_t}, \qquad g_t^2 = \dot{\sigma}_t^2 - 2f_t\sigma_t^2 = 2\alpha_t\sigma_t\,\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{\sigma_t}{\alpha_t}\Big). \tag{33}$$

This bijection $(\alpha_t, \sigma_t) \leftrightarrow (f_t, g_t)$ allows seamless translation between FM schedules and SDE coefficients.

10.3 VP and VE as special cases

Example 10.4 (Variance Preserving (VP))**.** Forward SDE: $\mathrm{d}Y_t = -\frac{1}{2}\beta(t)Y_t\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}W_t$, $Y_0 \sim p_{\text{data}}$. Let $\Gamma(t) = \int_0^t \beta(u)\,\mathrm{d}u$. Then $f_t = -\frac{1}{2}\beta(t)$, $g_t = \sqrt{\beta(t)}$, giving:

$$\alpha_t = e^{-\Gamma(t)/2}, \qquad \sigma_t^2 = 1 - e^{-\Gamma(t)}.$$

Note $\alpha_t^2 + \sigma_t^2 = 1$: this is the "variance preserving" property.

Example 10.5 (Variance Exploding (VE))**.** Forward SDE: $\mathrm{d}Y_t = g(t)\,\mathrm{d}W_t$, $Y_0 \sim p_{\text{data}}$. Here $f_t = 0$, $\alpha_t \equiv 1$, $\sigma_t^2 = \int_0^t g(u)^2\,\mathrm{d}u$.

11 Score identities, four parameterizations, and the FM = Diffusion equivalence

We now work under the affine Gaussian path and derive, in a single unified pass, all score-based identities and prediction equivalences. This section combines the score/posterior analysis with the four-parameterization framework, avoiding any duplication.

11.1 The affine Gaussian path

Assumption 11.1 (Affine Gaussian path — noise-to-data orientation)**.** Let $X_0 \sim \mathcal{N}(0, \mathrm{Id})$ (noise) and $Z \sim p_{\text{data}}$ (data) be independent. Let α_t, σ_t be C^1 schedules on $[0, 1]$ with

$$\alpha_0 = 0,\ \sigma_0 = 1, \qquad \alpha_1 = 1,\ \sigma_1 = 0, \qquad \sigma_t > 0 \text{ for } t \in [0, 1).$$

Define

$$\bar{X}_t \coloneqq \sigma_t X_0 + \alpha_t Z, \tag{34}$$

with density p_t and score $s(x, t) = \nabla_x \log p_t(x)$ for $t \in (0, 1)$.

Conditionally on $Z = z$, $\bar{X}_t \sim \mathcal{N}(\alpha_t z,\, \sigma_t^2\mathrm{Id})$. This is the same Gaussian conditional structure as (32), viewed in noise-to-data time. Both FM and diffusion operate on this shared structure.

11.2 Score marginalization for Gaussian mixtures

Lemma 11.2 (Score marginalization)**.** *Under Assumption 11.1, for $t \in (0,1)$:*

$$\nabla_x \log p_t(x) = \mathbb{E}\big[\nabla_x \log p_t(x \mid Z) \,\big|\, \bar{X}_t = x\big]. \tag{35}$$

Proof. From $p_t(x) = \int p_t(x \mid z)\, p_{\text{data}}(z)\, \mathrm{d}z$, differentiate under the integral:

$$\nabla p_t(x) = \int p_t(x \mid z)\, \nabla \log p_t(x \mid z)\, p_{\text{data}}(z)\, \mathrm{d}z.$$

Divide by $p_t(x)$ and recognize the posterior weighting $p(z \mid x) = p_t(x \mid z)\, p_{\text{data}}(z)/p_t(x)$. □

This is the score-level analogue of the marginalization trick (Proposition 5.1): the marginal score is a posterior expectation of conditional scores.

11.3 The four prediction types and their algebraic equivalence

Throughout, we write $x_t = \alpha_t x_0 + \sigma_t \varepsilon$ with $x_0 \sim p_{\text{data}}$, $\varepsilon \sim \mathcal{N}(0, \mathrm{Id})$, and $p_t(x_t \mid x_0) = \mathcal{N}(x_t;\, \alpha_t x_0,\, \sigma_t^2 \mathrm{Id})$.

The four predictions. All four aim to approximate a conditional expectation given the noisy observation x_t:

1. **ε-prediction** (noise): $\varepsilon^*(x_t, t) = \mathbb{E}[\varepsilon \mid x_t]$.
2. **x-prediction** (clean): $x^*(x_t, t) = \mathbb{E}[x_0 \mid x_t]$.
3. **Score prediction**: $s^*(x_t, t) = \nabla \log p_t(x_t)$.
4. **v-prediction** (velocity): $v^*(x_t, t) = \mathbb{E}[\dot{x}_t \mid x_t]$, where $\dot{x}_t = \alpha_t' x_0 + \sigma_t' \varepsilon$.

Proposition 11.3 (Algebraic equivalence of the four predictions)**.** *The four optimal predictions are related by:*

$$\varepsilon^*(x_t, t) = -\sigma_t\, s^*(x_t, t), \tag{36}$$

$$x^*(x_t, t) = \frac{1}{\alpha_t}\big(x_t + \sigma_t^2\, s^*(x_t, t)\big), \tag{37}$$

$$v^*(x_t, t) = \alpha_t'\, x^*(x_t, t) + \sigma_t'\, \varepsilon^*(x_t, t). \tag{38}$$

Proof. Equation (36)*.* The conditional score is $\nabla \log p_t(x_t \mid x_0) = -\frac{x_t - \alpha_t x_0}{\sigma_t^2} = -\frac{\varepsilon}{\sigma_t}$. By Lemma 11.2:

$$s^*(x_t, t) = \mathbb{E}\Big[-\frac{\varepsilon}{\sigma_t} \,\Big|\, x_t\Big] = -\frac{1}{\sigma_t}\varepsilon^*(x_t, t).$$

Equation (37)*.* From $x_t = \alpha_t x_0 + \sigma_t \varepsilon$:

$$x^* = \mathbb{E}[x_0 \mid x_t] = \frac{1}{\alpha_t}\big(x_t - \sigma_t\, \mathbb{E}[\varepsilon \mid x_t]\big) = \frac{1}{\alpha_t}\big(x_t + \sigma_t^2\, s^*\big),$$

using (36) to substitute $\varepsilon^* = -\sigma_t s^*$.

Equation (38)*.* The conditional velocity is $\dot{x}_t = \alpha_t' x_0 + \sigma_t' \varepsilon$. Taking the expectation given x_t yields $v^* = \alpha_t' x^* + \sigma_t' \varepsilon^*$ by linearity. □

Remark 11.4 (Connection to posterior means)*.* Equations (36)–(37) are equivalent to: $\mathbb{E}[X_0 \mid \bar{X}_t = x] = -\sigma_t\, s(x,t)$ and $\mathbb{E}[Z \mid \bar{X}_t = x] = \frac{1}{\alpha_t}(x + \sigma_t^2\, s(x,t))$ in the notation of Assumption 11.1 (where X_0 is noise and Z is data).

Their training objectives all share the template:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0,\,\varepsilon,\,t}\Big[\omega(t)\, \big\|\mathrm{NN}_\theta(x_t, t) - (A_t x_0 + B_t \varepsilon)\big\|^2\Big]. \tag{39}$$

Prediction type	A_t	B_t	**Regression target**
Clean (x-pred.)	1	0	x_0
Noise (ε-pred.)	0	1	ε
Score	0	$-1/\sigma_t$	$-\varepsilon/\sigma_t$
Velocity (v-pred.)	α_t'	σ_t'	$\alpha_t' x_0 + \sigma_t' \varepsilon$

Table 3: Regression target $= A_t x_0 + B_t \varepsilon$ for each parameterization.

11.4 The FM velocity is the PF-ODE drift: the complete equivalence

We now derive the central identity connecting FM and diffusion.

Theorem 11.5 (FM velocity = PF-ODE drift)**.** *Under Assumption 11.1, the FM optimal velocity* $v^*(x,t) = \mathbb{E}[\dot{\bar{X}}_t \mid \bar{X}_t = x]$ *satisfies*

$$v^*(x,t) = \frac{\dot{\alpha}_t}{\alpha_t}\, x + \sigma_t\Big(\frac{\dot{\alpha}_t}{\alpha_t}\sigma_t - \dot{\sigma}_t\Big) s(x,t). \tag{40}$$

Using the schedule conversion $f_t = \dot{\alpha}_t/\alpha_t$ *and* $g_t^2 = 2\sigma_t\dot{\sigma}_t - 2f_t\sigma_t^2$ *(Proposition 10.3), this is exactly:*

$$\boxed{v^*(x,t) = f_t\, x - \tfrac{1}{2} g_t^2\, \nabla \log p_t(x),} \tag{41}$$

which is the probability-flow ODE drift from Theorem 4.1.

Proof. From (34), $\dot{\bar{X}}_t = \dot{\sigma}_t X_0 + \dot{\alpha}_t Z$. Conditioning on $\bar{X}_t = x$ and using the posterior mean identities from Proposition 11.3:

$$\begin{aligned} v^*(x,t) &= \dot{\sigma}_t\, \mathbb{E}[X_0 \mid x] + \dot{\alpha}_t\, \mathbb{E}[Z \mid x] \\ &= \dot{\sigma}_t\, (-\sigma_t\, s) + \dot{\alpha}_t \cdot \frac{1}{\alpha_t}(x + \sigma_t^2\, s) \\ &= \frac{\dot{\alpha}_t}{\alpha_t}\, x + \Big(\frac{\dot{\alpha}_t}{\alpha_t}\sigma_t^2 - \dot{\sigma}_t\sigma_t\Big) s, \end{aligned}$$

which is (40). Now substitute $f_t = \dot{\alpha}_t/\alpha_t$:

$$v^*(x,t) = f_t\, x + \big(f_t\sigma_t^2 - \dot{\sigma}_t\sigma_t\big)\, s = f_t\, x - \frac{1}{2}\big(2\dot{\sigma}_t\sigma_t - 2f_t\sigma_t^2\big)\, s = f_t\, x - \frac{g_t^2}{2}\, s.$$

□

Gaussian FM = Diffusion: what this theorem means

Under affine Gaussian conditional paths:

1. The FM optimal velocity field *is* the probability-flow ODE drift — the same ODE used for deterministic sampling in diffusion models.
2. Training FM with v-prediction (CFM loss with target $\alpha_t' x_0 + \sigma_t' \varepsilon$) is gradient-equivalent to training diffusion with score/noise prediction, up to a time-dependent reweighting (since the four predictions are algebraically interchangeable by Proposition 11.3).
3. FM and diffusion are not two competing frameworks: they are the *same mathematical object* viewed through different lenses (conditional velocity fields vs. score functions).

11.5 PF-ODE in all four parameterizations

Using the inter-conversion formulas from Proposition 11.3, the PF-ODE $\dot{x} = v^*(x,t)$ admits equivalent forms depending on which network output is available.

Proposition 11.6 (PF-ODE forms)**.**

$$\begin{aligned}
\frac{\mathrm{d}x}{\mathrm{d}t} &= \frac{\alpha_t'}{\alpha_t}x - \sigma_t\Big(\frac{\alpha_t'}{\alpha_t} - \frac{\sigma_t'}{\sigma_t}\Big)\varepsilon^*(x,t) && (\varepsilon\text{-prediction}) && (42)\\
&= \frac{\sigma_t'}{\sigma_t}x + \alpha_t\Big(\frac{\alpha_t'}{\alpha_t} - \frac{\sigma_t'}{\sigma_t}\Big)x^*(x,t) && (x\text{-prediction}) && (43)\\
&= \frac{\alpha_t'}{\alpha_t}x + \sigma_t^2\Big(\frac{\alpha_t'}{\alpha_t} - \frac{\sigma_t'}{\sigma_t}\Big)s^*(x,t) && (\text{score prediction}) && (44)\\
&= v^*(x,t) && (v\text{-prediction}). && (45)
\end{aligned}$$

The simplest parameterization

In v-prediction, the PF-ODE is simply $\dot{x} = v^*(x,t)$: the velocity field *is* the ODE drift. This is why flow matching naturally trains a v-predictor, while diffusion traditionally trains an ε- or score-predictor and reconstructs the ODE drift via (42) or (44).

Example 11.7 (VP simplification)**.** Under VP schedules $\alpha_t = \sqrt{\bar{\alpha}_t}$, $\sigma_t = \sqrt{1-\bar{\alpha}_t}$, define $\kappa_t = \dot{\alpha}_t/\alpha_t = \dot{\bar{\alpha}}_t/(2\bar{\alpha}_t)$. Then the coefficient of s in (40) equals κ_t as well: $\sigma_t(\kappa_t\sigma_t - \dot{\sigma}_t) = \kappa_t$. The FM velocity simplifies to $v^*(x,t) = \kappa_t\,(x + s(x,t))$.

12 All affine flows are equivalent

Proposition 12.1 (Equivalence of affine flows)**.** *Let $x_t^{\mathrm{FM}} = (1-t)x_0 + t\varepsilon$ be the canonical FM interpolation and $x_t = \alpha_t x_0 + \sigma_t\varepsilon$ any affine interpolation. Define $c(t) \coloneqq \alpha_t + \sigma_t$ and $\tau(t) \coloneqq \sigma_t/(\alpha_t + \sigma_t)$ (assuming $c(t) \neq 0$). Then:*

$$x_t = c(t)\,x_{\tau(t)}^{\mathrm{FM}}, \tag{46}$$

and the velocity transforms as:

$$v(x_t,t) = c'(t)\,x_{\tau(t)}^{\mathrm{FM}} + c(t)\,\tau'(t)\,v^{\mathrm{FM}}\big(x_{\tau(t)}^{\mathrm{FM}}, \tau(t)\big). \tag{47}$$

Proof. Direct algebraic rewrite: $x_t = (\alpha_t + \sigma_t)\big(\frac{\alpha_t}{\alpha_t+\sigma_t}x_0 + \frac{\sigma_t}{\alpha_t+\sigma_t}\varepsilon\big) = c(t)\big((1-\tau(t))x_0 + \tau(t)\varepsilon\big) = c(t)\,x_{\tau(t)}^{\mathrm{FM}}$.

For the velocity, apply the chain rule to $x_t = c(t)\,x_{\tau(t)}^{\mathrm{FM}}$ and use $v^{\mathrm{FM}}(x,\tau) = \varepsilon - x_0$ (constant along the canonical path). □

Practical consequence

The "simplicity" of the linear FM schedule ($\alpha_t = 1-t$, $\sigma_t = t$) is not fundamental: any affine interpolation is equivalent up to time reparameterization and spatial rescaling. Different schedules affect training dynamics, weighting, and numerical properties, but not the class of representable flows.

Part V
Synthesis

13 The Fokker–Planck equation as the unifying principle

We conclude by summarizing the logical architecture of the course in a single diagram.

Unified perspective

All generative models in this course share one underlying rule: **the Fokker–Planck / continuity equation** governing the evolution of $p_t(x)$ under a chosen forward process.

Forward process (from data to noise):

- Variational: defining $p_t(x_t \mid x_0) = \mathcal{N}(\alpha_t x_0, \sigma_t^2 \mathrm{Id})$.
- SDE: $\mathrm{d}x = f(t)x\,\mathrm{d}t + g(t)\,\mathrm{d}w$, with $(f, g) \leftrightarrow (\alpha_t, \sigma_t)$ via Proposition 10.3.

Reverse process (from noise to data), all sharing the same p_t:

- PF-ODE: $\mathrm{d}x = v^*(x,t)\,\mathrm{d}t$ where $v^* = f(t)x - \frac{1}{2}g^2(t)\nabla \log p_t(x)$ $(\gamma \equiv 0)$
- Reverse SDE: drift corrected by $-g^2 s$ plus diffusion $g\,\mathrm{d}\bar{W}$ $(\gamma = g)$
- General SDE: $\mathrm{d}x = [v^* + \frac{\gamma^2}{2}s]\,\mathrm{d}t + \gamma\,\mathrm{d}B$ for any $\gamma(t) \geq 0$ (Thm. 4.4)

Learning:

- Score matching $\leftrightarrow$ Denoising score matching $\leftrightarrow$ Conditional flow matching.
- All are conditional expectations: $s^* = \mathbb{E}[\nabla \log p_t(x_t \mid x_0) \mid x_t]$, $v^* = \mathbb{E}[\dot{x}_t \mid x_t]$.
- Training with conditional targets differs from marginal targets by a θ-free constant.

Equivalences (Theorem 11.5 and Propositions 11.3, 12.1):

- Four parameterizations (ε, x, score, velocity) are algebraically interchangeable.
- All affine flows are equivalent up to time reparameterization and rescaling.
- Gaussian FM = Diffusion: the FM optimal velocity *is* the PF-ODE drift $f_t x - \frac{1}{2}g_t^2 \nabla \log p_t$.

A Derivation of weak forms from stochastic calculus

A.1 ODE case

If $\dot{X}_t = v_t(X_t)$ and $\varphi \in C_c^\infty$, the chain rule gives $\frac{\mathrm{d}}{\mathrm{d}t}\varphi(X_t) = \langle \nabla\varphi(X_t), v_t(X_t)\rangle$. Taking expectations yields the weak continuity equation (7).

A.2 SDE case

If $\mathrm{d}X_t = b(X_t, t)\,\mathrm{d}t + \sigma(t)\,\mathrm{d}B_t$, Itô's formula gives

$$\mathrm{d}\varphi(X_t) = \langle \nabla\varphi(X_t), b(X_t, t)\rangle\ \mathrm{d}t + \sigma(t)\,\langle \nabla\varphi(X_t),\, \mathrm{d}B_t\rangle + \frac{\sigma(t)^2}{2}\Delta\varphi(X_t)\,\mathrm{d}t.$$

Taking expectations (the Itô integral contributes zero) yields the weak Fokker–Planck equation (2).